

R Cheat Sheet: Continuous Distributions

1. R Distribution Naming Pattern

All probability distributions in R follow the same structure. Once you understand the pattern, you can use any distribution.

Prefix	Meaning	Example
d_	Density (height of curve)	<code>dnorm(x, mean, sd)</code>
p_	Probability $P(X < x)$	<code>pnorm(x, mean, sd)</code>
q_	Quantile / percentile (cutoff value)	<code>qnorm(p, mean, sd)</code>
r_	Random simulation	<code>rnorm(n, mean, sd)</code>

2. Computing Probabilities

p_ functions return cumulative probabilities:

$$F(x) = P(X < x)$$

Distribution	Probability Function	Connection to textbook parametrization
Normal	<code>pnorm(x, mean, sd)</code>	μ, σ
Uniform	<code>punif(x, min, max)</code>	θ_1, θ_2
Exponential	<code>pexp(x, rate)</code>	$\text{rate} = \lambda = 1/\beta$
Gamma	<code>pgamma(x, shape, rate)</code>	$\text{shape} = \alpha, \text{rate} = \lambda = 1/\beta$
Chi-square	<code>pchisq(x, df)</code>	ν
Beta	<code>pbeta(x, shape1, shape2)</code>	α, β

Common Probability Patterns / Tasks

Goal	R code pattern
$P(X < x)$	<code>p_(x, ...)</code>
$P(X > x)$	<code>1 - p_(x, ...)</code> OR <code>p_(x, ..., lower.tail = FALSE)</code>
$P(a < X < b)$	<code>p_(b, ...) - p_(a, ...)</code>

By default R gives you the LOWER tail probability (to the left of a cutoff, $P(X < x)$). If you want the upper tail probability $P(X > x)$, you have to do 1 minus the lower tail or add the argument `lower.tail = FALSE` inside the function.

4. Finding Percentiles (Cutoffs)

Use the `q_()` functions.

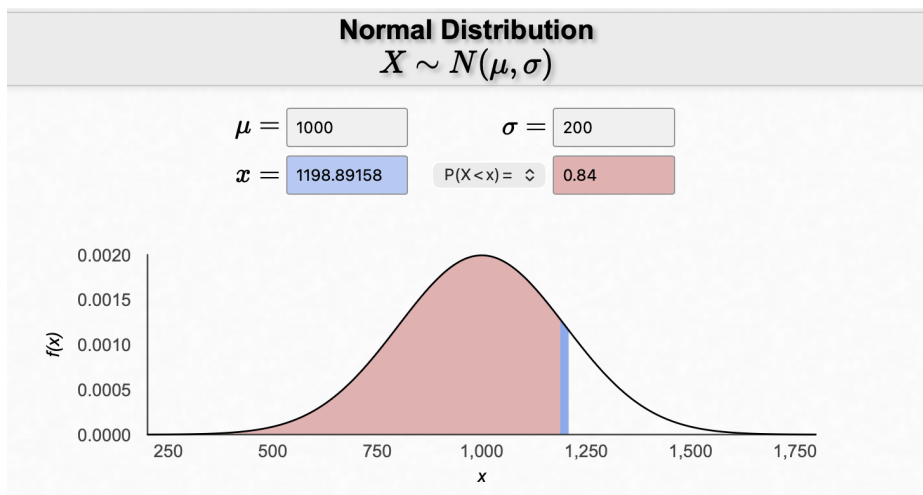
They find x such that:

$$P(X < x) = p$$

Example: `qnorm(.84, 1000, 200)` returns the 84th percentile for a normal distribution with mean $\mu = 1000$ and standard deviation $\sigma = 200$.

```
qnorm(.84, 1000, 200)
```

```
[1] 1198.892
```



Note, the `p_()` and `q_()` functions are inverses.

```
pnorm(1198.89158, 1000, 200)
```

```
[1] 0.84
```